

1 Autoplay on Trial: An Online Experiment on the Suspect 1
2 Behind Excessive Screen Time 2

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5 **Abstract** 5

6 Interface design can be strategically employed to ‘nudge’ consumers to take certain ac- 6
7 tions and is suspected of promoting addictive online behavior. A prevalent interface 7
8 element across numerous popular social media and streaming platforms is the default 8
9 autoplay feature. In this study, we present an incentivized online experiment to inves- 9
10 tigate whether the autoplay feature alone can cause an unwarranted increase in desired 10
11 viewing times. While controlling for content, we demonstrate that the autoplay feature, 11
12 in isolation, does not override participants’ previously indicated preferences for media 12
13 consumption. We discuss the our findings in terms of what actually renders online plat- 13
14 forms addictive. Our results have direct policy implications for the proposed blanket 14
15 ban on autoplay by the US Congress (SMART Act). 15

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1 Introduction

Autoplay is a user interface feature which automatically transitions users to new video content without any action. It has become a familiar design element in major social media and streaming platforms over the last decade,¹ and has been described as a default nudge that reduces the autonomy of users (Lukoff et al., 2021; Schaffner et al., 2025). Autoplay’s widespread adoption and concerns about its restricting the freedom of choice has also attracted regulatory scrutiny, with the U.S. SMART Act proposing to ban platforms from serving autoplay content “without an express, separate prompt by the user” due to suspicions of its contribution to addictive online behavior.²

While autoplay is pervasive and tends increase viewing times (Chen et al., 2025; Hiniker et al., 2018; Schaffner et al., 2025), its impact on user welfare and the mechanisms driving this effect are not well understood. Does the increase in consumption go against users’ wishes, or does it merely remove frictions to help them achieve their desired consumption levels? Can we attribute the increase in consumption to autoplay in isolation, or does it also depend on the supply of content? One reason for these gaps is the challenge of holding constant the video content displayed to users while eliciting their preferences related to content consumption before the actual consumption occurs.

We overcome this challenge in an incentivized two-day online experiment designed to isolate the causal effect of autoplay on content consumption. We recruited 184 participants who twice allocated 1200 seconds (20 minutes) between two tasks. Allocations were first made hypothetically using the strategy method, one day before the work date, and then realized on the work date. The tasks involved seconds of transcription of meaningless CAPTCHA texts at 0.38 pence/second, and seconds of watching funny animal videos at 0.33 pence/second. On the first day, participants familiarized themselves with

¹As of 2025 these platforms include Facebook, Instagram, Netflix, YouTube, TikTok, and Twitter (currently X). Twitter first introduced autoplay in 2015, and described it as a means to reduce “extra effort” in the number of clicks/taps required to consume content: <https://blog.x.com/official/en.us/a/2015/introducing-a-more-seamless-video-experience-with-autoplay.html>.

²SMART Act of 2019, S. 2314, 116th Congress: <https://www.congress.gov/bill/116th-congress/senate-bill/2314/text>

both tasks in a practice session and made their preferred time allocation for the next day. We incentivized these allocations by randomly implementing 5% of their choices on the work date. On the next day, remaining participants allocated their time without constraints. Differences between planned and realized allocations allowed for the measurement of dynamic inconsistency, standard in intertemporal choice experiments (Ericson & Laibson, 2018). This two-day structure with hypothetical and realized choices formed the foundation for testing our preregistered hypotheses regarding autoplay.

To isolate the causal effect of autoplay on these allocation decisions, we randomly assigned participants to either the **Autoplay** condition where videos played continuously during the watching task or to the **Control** condition requiring manual clicks to play each video. Across conditions, we held constant the content by presenting 80 manually curated “funny animal” videos from YouTube and TikTok in identical sequence. This eliminated algorithmic content curation and personalization as potential confounds. We dedicated the first block of the experiment to measure the differences between planned and realized allocations across conditions.

In a second block of the experiment, we modified the design to elicit demand for a commitment device. The commitment allowed participants to probabilistically favor their autoplay preference during 20 minutes on the main session. We recruited 52 additional participants who made nine binary choices between the autoplay setting (on or off) for the 20-minute main session and receiving bonus payments ranging from £0.05 to £0.50. One decision was randomly selected for implementation, determining both the autoplay setting for the main session and the associated bonus payment. This allowed us to measure participants’ Willingness To Pay (WTP) for autoplay in advance, providing insight into whether participants viewed the feature as beneficial (positive WTP) or as a self-control problem requiring a commitment device (negative WTP).

We proceed with reporting our main findings. The first set of results focuses on our preregistered hypotheses regarding the increase in content consumption that we attribute to autoplay. In the main session, participants in the **Autoplay** condition watched statistically indistinguishable numbers of videos compared to the **Control** condition,

and the **Autoplay** condition had no significant effect on the proportion of time spent on either task. Results also hold for other engagement metrics (i.e., average session length, number of sessions). To sum, we find no evidence that autoplay alone increased content consumption in our setting.

Second, contrary to our hypothesis that participants would demand commitment devices against autoplay, we find a positive WTP for the feature. Using data from the second block, we are able to estimate an average WTP of 2.24 pence (about 45 seconds of typing) per 20-minute session. This implies participants viewed autoplay as a convenience feature rather than as a threat to their self-control which would have required a costly commitment device.

Third, comparing hypothetical time allocations to realized ones, we find that participants consumed significantly less content than their day 1 plans implied in both conditions. The decrease amounts to 15.8 percentage points (35%) less time spent on the watching task with day 1 plans as the baseline. This result is driven by 33% of all participants who had planned to spend up to three quarters of their time watching videos but instead decreased their content consumption on the next day. On the other end, 88% of those who had initially planned to spend less than a quarter of their time watching videos successfully followed their plans and did not succumb to temptation.

Evidence from a regression discontinuity analysis supports that participants monitored their day 1 plans despite ultimately exceeding them. When participants reached their planned typing duration, they immediately reduced typing probability by 25.5 percentage points in the next 60-second window. However, this effect dissipated quickly, becoming insignificant within 90 seconds, as participants returned to the typing task once again.

Taken together, these findings cast doubt on whether the watching task was actually perceived as tempting. A mixed-method analysis of open-ended responses reveals that participants perceived the experimental environment as a work setting rather than a choice between independent tasks. While 29% spontaneously described typing using negative terms, another 23% characterized video-watching as “taking breaks” from work.

This suggests that experimenter demand effects may have interfered with the intended effects of the autoplay manipulation, with implications for both the interpretation of our null results and the broader generalizability of our study.

A primary internal validity concern is that experimenter demand effects favored allocating more time on the typing task than planned. These can emerge through contextual cues even without explicit behavioral instructions, as participants interpret the unfamiliar experimental environment by drawing inferences from the information provided, and may alter their behavior simply by having attention drawn to variables of interest (Zizzo, 2010). In our experiment, these could be (i) the small earnings difference favoring the typing task,³ (ii) multiple mentions of the quality criteria for the typing task to receive earnings, (iii) placement of the typing task as the first activity upon starting a session. Alternative designs could have addressed these concerns by rewarding balanced time allocations more, or by eliminating earning differences altogether.

Another major challenge in isolating the effects of autoplay is the role of content appeal. Higher content appeal would make the watching task more tempting, and create a stronger tradeoff between the two tasks. To address this while keeping the content identical across conditions, we conducted pretests from the same participant pool where the funny animal videos came out as the modal theme. In addition, participants rated the curated videos positively (median 7/10) in the post experiment survey. Regardless, the watching task lacked what participants could afford in terms of leisure - they could work harder for maximum earnings and wait 20 minutes to access personalized content privately. Allowing participants to select among a list of themes with manually curated videos could have addressed this issue (Ek & Samahita, 2023).

A final concern in online experiments is ensuring genuine participation equivalent to laboratory conditions. We implemented several measures to address this challenge. We used JavaScript-based attention monitoring to detect each second participants navigated away from the experimental interface following (Purohit et al., 2023). We also required

³This difference is 60 pence over 20 minutes, equivalent to a 1.8 pound hourly wage difference (15% above our 12 pound minimum wage baseline).

125 minimum internet speeds and restricted participation to desktop users with Chromium- 125
126 based browsers to standardize the user experience. Participants also answered post- 126
127 experiment survey questions about their engagement while watching videos and about 127
128 connection interruptions. These controls ensured that our final dataset is composed of 128
129 participants who were active, and approximate the controlled conditions of laboratory 129
130 experiments ([Arechar et al., 2018](#)). 130

131 We contribute to several strands of literature. First, the literature on studying 131
132 the effects of social media and streaming services on user well-being ([Allcott et al.,](#) 132
133 [2020, 2022](#); [Bao, 2025](#); [Beknazar-Yuzbashev et al., 2025](#); [Braghieri et al., 2022](#); [Bursztyn](#) 133
134 [et al., 2023](#); [Collis & Eggers, 2022](#); [Groshek et al., 2018](#); [Hoong, 2021](#); [Nyhan et al.,](#) 134
135 [2023](#); [Purohit et al., 2023](#); [Walton-Pattison et al., 2018](#)). We contribute to this work 135
136 by shedding light on one potential driver of the harmful effects of social media and 136
137 streaming services: the autoplay. 137

138 A subset of this literature focuses on the effects of specific interface design elements 138
139 on user behavior, particularly related to user agency and overconsumption of content 139
140 ([Bongard-Blanchy et al., 2021](#); [Chen et al., 2025](#); [Lukoff et al., 2021](#); [Lupiáñez-Villanueva](#) 140
141 [et al., 2022](#); [Lyngs et al., 2019, 2020](#); [Mathur et al., 2021](#); [Schaffner et al., 2023, 2025](#); 141
142 [Silverman et al., 2024](#)). To our knowledge, our paper provides the first attempt isolating 142
143 the causal effects of autoplay on content consumption. 143

144 Methodologically, this paper contributes to the literature by introducing an experi- 144
145 mental platform that directly records participant behavior and manipulates interface in 145
146 an online setting. Similar approaches using browser extensions XX 146

147 The remainder of this paper is organized as follows. Section 2 describes the experi- 147
148 mental design. Section 3 details our sample and procedures. Section 4 presents our 148
149 results and Section 5 concludes. 149

150	2 Design	150
151	2.1 Overview	151
152	To explore the relationship between default video player settings and self-control, we	152
153	designed a choice environment that captures the tension between productive effort and	153
154	immediate gratification. Our experimental setup is centered around two competing tasks:	154
155	A primary typing task that is tedious and offers limited learning opportunities, and a	155
156	secondary video-watching task that provides immediate entertainment value. This de-	156
157	sign mirrors real-world situations where individuals must allocate time between effortful	157
158	productive activities and readily available leisure alternatives.	158
159	The key feature in our design is the manipulation of autoplay setting in the video	159
160	interface. Participants can switch between tasks at any time, but the ease of continuing	160
161	with the leisure activity varies systematically across treatment conditions. This allows us	161
162	to isolate the causal effect of interface design on time allocation decisions while holding	162
163	constant individual time preferences and other task characteristics.	163
164	2.2 Timeline and Session Structure	164
165	We conducted a two-day online experiment to examine how autoplay features affect time	165
166	allocation between productive and leisure tasks. The multi-day structure allowed us to	166
167	measure the gap between stated preferences and actual behavior under two different	167
168	video interface conditions. Participants completed two separate data collection sessions	168
169	with a cooling-off period of 24 hours (up to 48 hours) between sessions.	169
170	Day 1 sessions included a practice session to familiarize participants with the exper-	170
171	imental interface, followed by preference elicitation measures. Day 2 featured the main	171
172	20-minute work session where participants made actual time allocation decisions. We	172
173	conducted two waves of data collection during May 2023, as described in Table 1.	173

Table 1: Summary of experimental design and timeline

	Practice Session	Time Choice	Main Session	Multiple Price List	Decision Horizon
Day 1 (May 10)	✓	✓			24-hour
Day 2 (May 11)			✓		immediate
Day 1 (May 18)	✓	✓		✓	24-hour
Day 2 (May 19)			✓		immediate

Note: This table shows the experimental components administered across two data collection waves, with columns ordered chronologically within each day. The first block (May 10-11 2023) established baseline treatment effects, while the second block (May 18-19 2023) incorporated the Multiple Price List treatment to measure demand for commitment devices. Decision horizon refers to the time between hypothetical preference elicitation and the main experimental session.

2.3 Payment Structure and Incentives

We implemented a performance-based payment system to ensure participants faced meaningful trade-offs between tasks. We based payment directly on the seconds participants actively spent on each task, with different compensation rates for the typing and video-watching activities. This created clear opportunity costs for time allocation decisions while avoiding any completion bonuses that might distort behavior.

Participants who completed all required elements of the experiment received payment within seven days after the second session.⁴ To maintain data quality, participants who failed to meet specified quality criteria for the typing task were not compensated for time spent on that task.

⁴Prolific policy gives researchers 21 days to transfer funds to participants. The typical delay between study completion and payment is however only a couple of days on average, as indicated on the Prolific website ([link](#)).

184 2.4 Technical Implementation and Interface Design 184

185 We designed the experimental platform to provide precise control over the testing envi- 185
186 ronment while maintaining ecological validity. We presented both tasks within a single 186
187 web page using distinct tabs, allowing participants to switch between activities seam- 187
188 lessly while preserving progress. The typing task always served as the default upon 188
189 session initiation, requiring an active choice to switch to the video-watching alternative. 189

190 Our interface incorporated several key features to track behavior precisely. A progress 190
191 bar and timer provided real-time feedback on session duration. We logged all critical 191
192 interactions including tab switches, keystrokes during typing, video controls, and cursor 192
193 movements outside the experimental window. This granular tracking enabled us to 193
194 construct detailed measures of attention and engagement across tasks. 194

195 2.5 Internal Validity Controls 195

196 Online experiments present unique challenges for maintaining experimental control, in- 196
197 cluding participant multitasking, technical disruptions, and environmental variation. We 197
198 implemented comprehensive measures to address these concerns and ensure data quality. 198

199 **Participant and Technology Screening:** We selectively admitted participants 199
200 using multiple screening criteria to ensure stable experimental conditions. Internet speed 200
201 requirements filtered out participants with unstable connections (minimum 30 Mbps on 201
202 day 1, 25 Mbps on day 2). We restricted participation to users with Windows, Linux, 202
203 or Mac operating systems using Chromium-based browsers exclusively. We excluded 203
204 Safari and Mozilla Firefox due to restrictive autoplay configurations. Mobile devices 204
205 and tablets were prohibited to standardize the user experience. 205

206 **Platform Controls:** The experimental website enforced linear navigation, prevent- 206
207 ing participants from moving backward through the experiment. We disabled right-click 207
208 functions and keyboard shortcuts. Any attempt to navigate backward reset progress 208
209 to the landing page. We clearly communicated these restrictions to participants at the 209
210 experiment’s outset. 210

211 **Attention and Engagement Monitoring:** We implemented real-time tracking 211

of participant attention using JavaScript packages to detect interface visibility disruptions.⁵ This system captured actions such as minimizing the browser, switching between web pages, or launching other applications. The tracking provided second-by-second accountability of participant engagement.

Post-Experiment Validation: After completing the experiment, participants reported any connectivity disruptions and confirmed their genuine engagement with the video-watching task. We combined these self-reports with our objective tracking measures to identify and filter data potentially compromised by distractions or divided attention.

2.6 Tasks

The experiment centers on participants' time allocation decisions between two distinct activities that differ substantially in their cognitive demands and entertainment value. Both the Typing and Watching Tasks were presented to participants twice: First during the Practice Session to familiarize participants with the interface and task mechanics, and second during the Main Session where performance determined compensation. In both sessions, depending on the randomized assignment to conditions, the key experimental manipulation occurs in the Watching Task.

2.6.1 Typing Task

The Typing Task requires retyping successive CAPTCHAs. CAPTCHAs are computer-generated images that contain random uppercase and lowercase letters as well as randomly placed white spaces.⁶ Each CAPTCHA has a total length of 35 characters, including the white spaces.

We designed the images to be blurred and distorted by lines and dots added on top to increase comprehension difficulty.⁷ The interface always presents these CAPTCHA

⁵For more information about the package, see [here](#).

⁶White spaces were added to improve readability after pretests. The number of white spaces was limited to a maximum of 5 instances per CAPTCHA to control the typing time per image.

⁷The randomization procedure to generate CAPTCHAs is available [here](#).

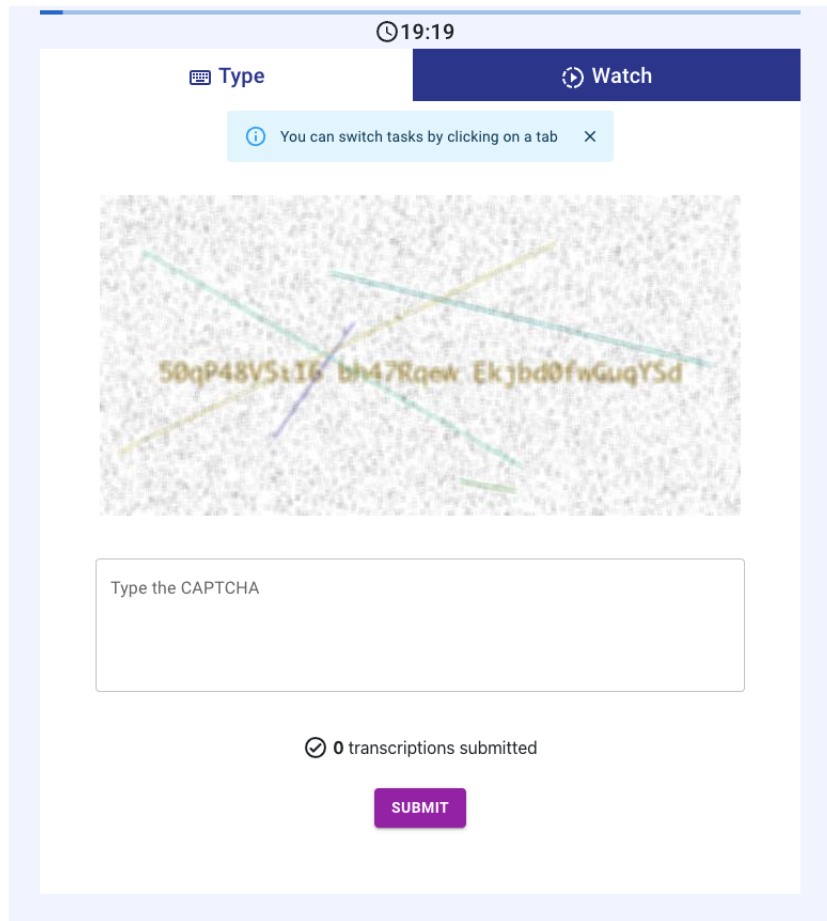
236 images with a text box and a submit button. When the submit button is clicked, the 236
237 next CAPTCHA image appears and the text input for the previous one is stored. The 237
238 task is identical across all conditions, as illustrated in Figure 1. 238

239 We established easy-to-meet quality requirements to ensure participants remained 239
240 active. These requirements include an overall typing accuracy of at least 70 percent 240
241 in submitted CAPTCHAs⁸ and a minimum of one CAPTCHA submission per minute 241
242 spent on the task. 242

243 Participants must meet these quality requirements to receive compensation for the 243
244 Typing Task. We remunerated the Typing Task at 0.38 pennies per second participants 244
245 worked on it. A hidden timer recorded each second participants kept the Typing Task 245
246 in the visible part of their screen. Participants in both conditions spent on average 41 246
247 seconds per CAPTCHA. 247

⁸We use Python’s `difflib` string-matching algorithm to measure accuracy. It consists of finding the longest common substring and then finding recursively the number of matching characters in the non-matching regions on both sides of the longest common substring. Overall accuracy of 70 percent implies an average of 10.5 ($0.3 * 35$) mistakes across all submitted CAPTCHAs.

Figure 1: Typing Task tab



Note: This figure describes the typing interface tab. As participants always begin in the “Type” tab by default, there is a banner that reminds participants to click the tab buttons to switch tasks. This banner disappeared as soon as participants switched tasks once.

2.6.2 Watching Task

In this task, participants watch successive short videos in a customized media player. Since we are interested in isolating the media player’s default settings, the video content and the order in which videos appear to participants are identical across conditions. We completely disabled the media player controls: Participants cannot skip videos (user scrolling is disabled) or use the media player’s slider to advance or go back in a particular

254 video. We remunerated the Watching Task at 0.33 pennies per second participants 254
255 worked on it. 255

256 The conditions differ in their autoplay functionality. In the **Autoplay** condition, 256
257 videos play automatically when the Watching tab is visible and pause with a click any- 257
258 where on the video. While the Watching task is visible, the media player keeps scrolling 258
259 to the next videos and continues playing them unless interrupted. In the **Control** con- 259
260 dition, participants need to have the Watching tab visible and click on each consecutive 260
261 video to play them. At the end of each video, the scrolling animation brings the next 261
262 video, but the new video only starts playing when the participant clicks on it. Figure 2 262
263 illustrates the video tab in the **Control** condition. 263

264 We curated videos from YouTube and TikTok, using tags such as “funny animal 264
265 videos/short” and “cute animal videos/shorts”. We manually selected 80 videos to iso- 265
266 late the difference in the media player design and help avoid algorithmic biases resulting 266
267 from the recommender system’s personalized suggestions. We chose animal videos be- 267
268 cause of their popularity during pilot studies.⁹ We reviewed the selected videos to ensure 268
269 that they did not contain any harmful or violent actions toward the animals involved. 269
270 We kept the first 60 videos shorter in the beginning to maximize the frequency of expe- 270
271 riencing the autoplay feature during the watching session, with videos getting longer for 271
272 the last 20 videos (see Table 2). 272

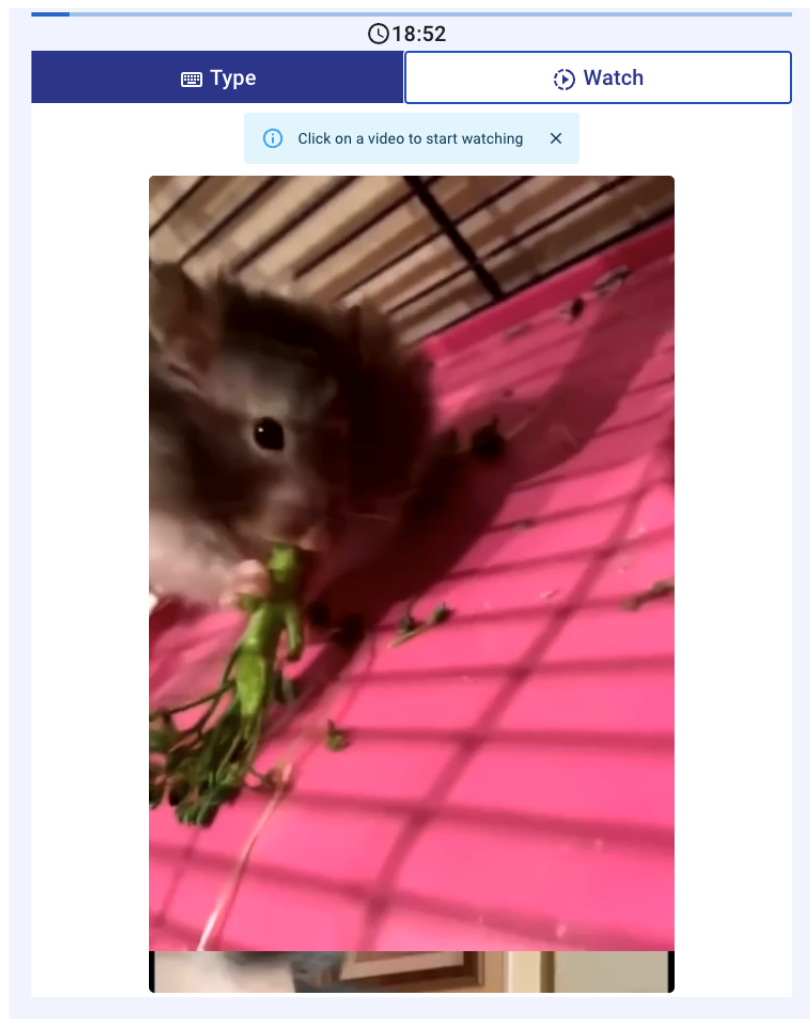
⁹During the pilots, the experiment was followed by a separate questionnaire asking participants to rate the videos they had seen and suggest what types of videos they would have liked to see more of.

Table 2: Video durations in the watching task

	First 60 videos	Last 20 videos
<i>Duration (seconds)</i>		
Mean (SD)	12.25 (7.04)	25.50 (14.04)
Median [Min, Max]	10 [5, 38]	19.5 [8, 60]

Note: This table presents the duration characteristics of the 80 animal videos used in the experiment. The first 60 videos were kept shorter to maximize participants' exposure to the autoplay feature during watching sessions, while the last 20 videos were longer.

Figure 2: Watching Task tab



Note: This figure describes the “Watch” tab in the **Control** condition. Before the first video is clicked upon to be played, there is a banner that reminds participants that the video won’t play without clicking. This banner was not present in the **Autoplay** condition and videos started playing as soon as participants clicked on the “Watch” button seen up on the right. Notice how top part of the following video is already visible in the bottom part of the screen.

2.7 Practice Session

The experiment consists of two sessions during which participants interact with the tasks described above. The first session is the Practice Session on day one. It comes right

after the written descriptions of each task and serves to familiarize participants with the tasks and the interface. Participants are assigned to either **Autoplay** or **Control** condition before the Practice Session, so that they experience the same video interace on both the Practice and Main Sessions.

During the practice session, each task was available for only 60 seconds and participants were not allowed to switch between tasks. On the first tab, participants were required to retype CAPTCHAs. The 60-second period was enough to submit one CAPTCHA and see the second CAPTCHA image. After 60 seconds, a pop-up appeared on the screen blocking any other interaction and taking participants to the second page.

Participants were required to close this pop-up to advance to the second task. The rationale behind the pop-up was to ensure participants spent equal time on both tasks. Whether or not this pop-up was closed was verified to check participant activity and served as an additional attention check.¹⁰ Once closed, the Watching Task began for 60 seconds, enabling participants to watch 4 to 5 short videos depending on whether they were assigned to the **Autoplay** or the **Control** condition. In the **Autoplay** condition, videos played continuously, while in the **Control** condition, videos required manual start.

We did not remunerate participants for the time they spent here because the purpose of the practice session was to understand the difficulty and pacing of the tasks, as well as navigation within the interface.

2.8 Multiple Price List

After the Practice Session on day one, participants in the Multiple Price List (**MPL**) condition made 9 binary decisions (see Figure 3). They chose between their preferred media player setting (Autoplay off or on) and a bonus payment. We introduced the MPL condition after the initial data collection phase. We determined its bonuses by the per-second earning differences across tasks, multiplied by the observed time differences between the Autoplay and control groups.

¹⁰All participants closed this pop-up and switched to the Watching Task.

Specifically, the average difference between the Autoplay and control treatments, in terms of time spent across tasks, was approximately 60 seconds (as shown in Table 8). This resulted in an average earning difference of $60 * (0.38 - 0.33) = 3$ pennies. To capture this in the MPL choices offered, we set the symmetric bonuses accordingly: £[0.05, 0.1, 0.25, 0.5].

Once these decisions were completed, participants received information on the media player setting for day two—specifically whether Autoplay would be on or off—and the associated bonus. This information was displayed on the next page. With this knowledge, they then decided on their Time Choice. Participants who were not in the MPL condition proceeded directly to the Time Choice page after their Practice Session. More details on the Time Choice can be found in the subsequent section.

Figure 3: Multiple Price List page

Autoplay on or off?

In the practice session videos played automatically with **Autoplay**. There is another version **where you need to click on videos to play them**.

You need to make 9 decisions about Autoplay. One of your decisions will be randomly chosen and implemented. **You will receive a bonus payment and watch videos with the chosen Autoplay setting.**

For example, if you choose Autoplay for the first decision and it is implemented, you will have Autoplay and receive an additional £0.5 bonus payment.

▶ AUTOPLAY +£0.5	⏮ NO AUTOPLAY +£0
▶ AUTOPLAY +£0.25	⏮ NO AUTOPLAY +£0
▶ AUTOPLAY +£0.1	⏮ NO AUTOPLAY +£0
▶ AUTOPLAY +£0.05	⏮ NO AUTOPLAY +£0
▶ AUTOPLAY +£0	⏮ NO AUTOPLAY +£0
▶ AUTOPLAY +£0	⏮ NO AUTOPLAY +£0.05
▶ AUTOPLAY +£0	⏮ NO AUTOPLAY +£0.1
▶ AUTOPLAY +£0	⏮ NO AUTOPLAY +£0.25
▶ AUTOPLAY +£0	⏮ NO AUTOPLAY +£0.5

⚠ Please pick your favorite option for each row.

CONTINUE

Note: This figure describes the MPL page where participants made 9 binary choices between their preferred autoplay setting and bonus payments. Each row presents a choice between “Autoplay” (left) and “No Autoplay” (right) with associated bonus amounts ranging from £0.05 to £0.5. The interface required participants to select one option in each row before proceeding. One of these 9 decisions was randomly selected for implementation in the Main Session, determining both the autoplay setting and any bonus payment for day two.

314

2.9 Time Choice

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Following the Practice Session, participants transitioned to the Time Choice section.

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315

This section required them to specify their preferred time allocation between the two

316

317 tasks for the main 20-minute session scheduled for the following day. Participants knew 317
318 the main session would be 20 minutes long and they were free to choose corner solutions, 318
319 even if that meant focusing solely on one task. 319

320 We designed a slider tool with 20-second steps for participants to indicate their time 320
321 allocation preferences. Initially, the slider appeared grayed out and required a click 321
322 to activate. This design aimed to ensure participants considered their choices in an 322
323 active manner. The click-to-activate design had two benefits: It guaranteed participants 323
324 interacted with the slider and helped participants understand the task-related payoff 324
325 structure. 325

326 To emphasize the importance of this decision, we made the choices of 5 percent of 326
327 participants binding in the main session. Binding choices led to payments based on 327
328 the initial Time Choice, but only if participants met the quality standards described 328
329 under Section 2.6.1. We excluded data from these participants from the final dataset 329
330 accordingly. We informed participants whether their allocation was binding with their 330
331 Time Choice reminder on day two, right before the start of the main session. 331

332 The Time Choice served as a benchmark for participants, and participants accord- 332
333 ingly received a textual reminder of it on day two. The Time Choice section offers 333
334 insights into participants' informed time preferences allows gauging their commitment 334
335 to these choices after the 24-hour cooling period. 335

Figure 4: Time Choice page

Time Choice

Now decide how long you would like to spend **tomorrow** on **Typing** and on **Watching Videos**.

Tomorrow you will have **20 minutes** for both tasks.

1 out of every 20 participant will be **bound** by the choice they made today.

Please **click on the slider** and indicate your **Time Choice**:

i You choose to spend **10 minutes 0 seconds** to **Type**.
 You choose to spend **10 minutes 0 seconds** to **Watch Videos**.
 You would earn **£4.25**.

Tomorrow you will learn whether your **Time Choice** is binding.

CONTINUE

Note: This figure shows the Time Choice interface with a slider tool allowing participants to allocate time between the two experimental tasks for the following day’s 20-minute Main Session. The slider initially appears grayed out and requires a click to activate. Real-time feedback shows the selected allocation (here, 10 minutes each) and expected earnings (£4.25). The interface informs participants that 1 out of 20 (5 percent) will be bound by their choice and that they will learn whether their choice is binding on day two.

336 2.10 Main Session 336

337 The Main Session took place on day two, approximately 24 hours after the Time Choice. 337
 338 We began by reminding participants of their previous day’s allocation and informing 338
 339 them whether their choice was binding or whether they were free to allocate their time 339
 340 as they preferred. 340

341 The session lasted exactly 20 minutes. Participants could freely switch between the 341
 342 Typing and Watching Tasks at any time using the tab interface. Unlike the Practice 342
 343 Session, there were no time restrictions on individual tasks, and participants had com- 343

plete autonomy over their time allocation decisions. The interface maintained real-time tracking of time spent on each task, and participants could monitor their progress and potential earnings through the progress bar and timer.

We implemented the experimental treatment during this session through the media player’s autoplay functionality. Participants assigned to the **Autoplay** condition experienced automatic video playback when the Watching tab was visible, while those in the **Control** condition required manual clicks to start each video. All other task mechanics remained identical to the Practice Session.

Payment was calculated based on actual time spent on each task, with compensation rates of 0.15 pence per second for Typing and 0.1 pence per second for Watching. Participants needed to meet the established quality requirements for the Typing Task to receive compensation for time spent on that activity. The session concluded automatically after 20 minutes, and participants received their final payment calculation before completing the experiment. Participants received a fixed completion fee of £2.75. Average earning per participant was £4.08, amounting to an average payment of £1.33 (median £1.49) for the main session.

2.11 Hypotheses

[Kleinberg et al. \(2022\)](#) model user consumption as driven by two competing systems: System 1 (impulsive, myopic) and System 2 (thoughtful, forward-looking). System 2 represents users’ “actual” preferences, while System 1 drives continued consumption even when users derive little utility. Content exists within a *content manifold* that can be characterized along three dimensions: how much users genuinely value the content, how long they naturally want to engage with it, and how “sticky” or addictive it is akin to the difference between nutritious salad and moreish potato chips.

Following this framework, autoplay reduces friction in content consumption by removing decision points between videos, making content more “sticky” and decrease the likelihood that users’ thoughtful system will regain control between consumption decisions. The key assumption here is that the experimental videos have sufficient baseline

appeal and stickiness to allow autoplay to meaningfully increase engagement. As [Kleinberg et al. \(2022\)](#) demonstrate, UI design choices are “not separable from other content choices, since their effect depends on where the platform has positioned its content on the underlying content manifold.” If content is purely “salad-like” with very low inherent appeal and stickiness, autoplay may have minimal effect on video consumption. This is analogous to how introducing breaks would have little impact on engagement for low-stickiness content like documentaries, but dramatically reduce engagement for high-stickiness content like celebrity gossip. With the assumption that our selection of videos have sufficient appeal and stickiness, **Autoplay** condition should lead to consumption beyond users’ stated preferences, creating a measurable gap between intended (Time Choice) and actual behavior. In this case, we expect autoplay’s behavioral effects to be known by participants, albeit underestimated as to how much they would be affected by these interfaces ([Bongard-Blanchy et al., 2021](#)). We thus expect positive demand for a commitment device driven by “sophisticated” participants in line with time preference literature.

Hypothesis 1: **Autoplay** condition decreases time allocated to the typing task while increasing video consumption. We test this using a linear regression:

$$Y_i = \alpha + \beta \text{Autoplay}_i + \gamma \text{TimeChoice}_i + \delta(\text{Autoplay}_i \times \text{TimeChoice}_i) + \epsilon_i \quad (1)$$

where Y_i represents our outcome variables of interest (proportion of time spent typing or number of videos watched), Autoplay_i is a binary indicator for the treatment, and TimeChoice_i is the standardized proportion of time participants planned to spend typing on day 1. We estimate this model with and without the interaction term to examine whether the autoplay effect varies based on initial planning preferences. We expect $\beta < 0$ for typing and $\beta > 0$ for videos watched.

Hypothesis 2: **Autoplay** condition has an effect on deviations from stated time allocation preferences (actual minus planned time spent on typing). We test this using a two-sample t -test comparing the means between the two groups.

Hypothesis 3: If participants are aware of their self-control issues, they should

exhibit positive Willingness To Pay (WTP) for removing autoplay, preferring the commitment devices that restricts automatic video transitions. We test this using a logistic regression:

$$\Pr\left(\frac{\text{Autoplay}_{ij}}{1 - \text{Autoplay}_{ij}} | \text{Bonus}_{ij}\right) = \Lambda(\alpha + \beta \text{Bonus}_{ij}) \quad (2)$$

where Autoplay is the binary choice of taking autoplay given the bonus j from the MPL for individual i . We calculate the WTP for autoplay as $-\alpha/\beta$, and expect a positive sign indicating preference for commitment against autoplay.

In sum, these hypotheses test whether interface design features can systematically distort optimal time allocation decisions in a setting where participants face monetary incentives favoring productive work.¹¹

3 Sample and Procedures

The experiment was run entirely online, using widely available open-source tools.¹² We drew the sample from the Prolific participant pool, an online recruitment platform commonly used for academic research, targeting adult British residents. We further imposed the following criteria: Fluency in English, balanced gender representation, and having a stable internet connection (> 30 Mbps on the first and > 25 Mbps on the second day of the experiment). We did not allow mobile devices and restricted access to the experiment to Chromium-based browsers to enhance the internal validity of the experiment as previously discussed in Section 2.5. Only participants who succeeded on both attention checks, reported not having engaged in other activities, and did not have connection issues during the experiment were included in the final dataset.

For the first stage of data collection, that is to say excluding the MPL treatment, we preregistered a two-sample parallel design using our prototype data and tested for equality of means across treatments. We found that the required sample size to test

¹¹To access the preregistered hypotheses, click [here](#). Our preregistration included hypotheses 1 and 3. We have added supporting hypothesis 2 regarding our expectations about the Time Choice section later.

¹²We used the [MERN stack](#) to develop and host the website in an entirely free way.

Hypothesis 1 with the usual parameter values ($\alpha = .05$, $\beta = .2$) would be 114 per treatment, with an effect size of 65 seconds calculated by the difference in the average time spent between conditions ($\mu_{Autoplay} - \mu_{Control}$) and population standard deviation of 165 seconds. Since attrition in longitudinal experiments is commonplace, we invited 301 participants on the first day. On day two, we invited the same 301 participants and received 276 complete responses. No participants failed the attention checks more than twice, and therefore were not removed from the dataset as per Prolific policy. We dropped 17 participants who self-reported having connection issues and 24 additional participants who stated having engaged in other activities. We also removed 11 participants who had their Time Choice enforced randomly. Finally, we removed 40 participants that were detected to spend more than 20 consecutive seconds away from our experimental platform who disproportionately come from the autoplay condition. These 40 participants all reported engaging with videos but had their mouse detected outside the experiment window, making it impossible to distinguish whether they genuinely watched videos or browsed other tabs/windows during the study (see Appendix Table A.1). The final dataset on which we base the analysis consisted of 184 individuals. Table 3 shows the demographic characteristics of the observed individuals.

Table 3: Balance table for the sample

	Autoplay N = 79	Control N = 105	<i>p</i> [A = C]
<i>Gender</i>			
Female	48 (46.15%)	54 (45.00%)	0.510
<i>Age</i>			
Mean (SD)	40.56 (11.66)	40.24 (10.98)	0.725
<i>Employment</i>			
Employed (full time)	62 (59.62%)	67 (55.83%)	0.718
Employed (part time/self)	28 (26.92%)	28 (23.33%)	0.417
Not employed	11 (10.58%)	18 (15.00%)	0.167
<i>Income</i>			
Low (0-15)	28 (26.92%)	34 (28.33%)	0.989
Middle (15-30)	30 (28.85%)	36 (30.00%)	0.505
High (50+)	15 (14.42%)	13 (10.83%)	0.325
<i>Marital Status</i>			
Married/Partnership	56 (53.85%)	60 (50.00%)	0.510
Single	42 (40.38%)	52 (43.33%)	0.534
Previously married	6 (5.77%)	8 (6.67%)	0.927

Note: This table presents demographic characteristics for the sample of 184 participants across treatment conditions. Income categories are presented in thousands of British pounds (£). Employment categories: “Not employed” includes unemployed (looking and not looking) and homemaker. “Previously married” includes divorced, widowed, and separated. *p*-values for binary outcomes are from two-sample tests of proportions; for continuous variables, from two-sample *t*-tests. For employment, part-time and self-employed are together as one category. The sample demonstrates successful randomization with balanced participant characteristics across the conditions.

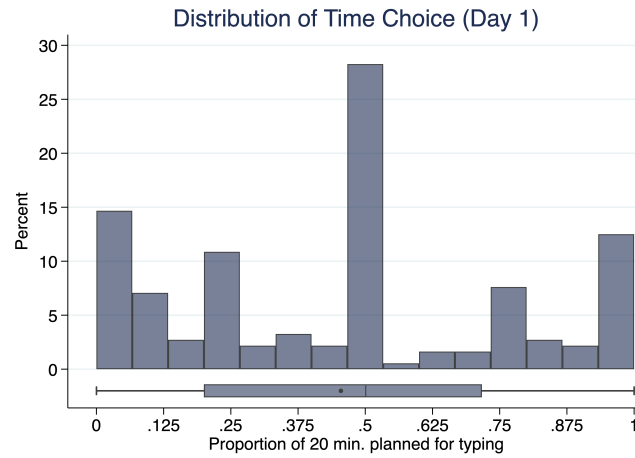
440 60 participants on day one. On day two, we received 57 complete answers. Once again, 440
441 no participant failed the attention checks. No participant reported having connection 441
442 issues. This improvement is likely because we had implemented higher network speed 442
443 requirements (> 35 Mbps on the first and > 30 Mbps on the second day). We dropped 443
444 5 participants who reported having engaged in other activities. As we were interested 444
445 in the demand for a commitment device, participants did not have their Time Choice 445
446 enforced in this treatment. We therefore ended up with 52 individuals in the final MPL 446
447 dataset. 447

448 4 Results 448

449 4.1 Time Choice: Planned time allocation on day 1 449

450 How did participants plan to allocate their time on day 1? We look at the participants' 450
451 Time Choices to answer this question. These were their incentivized time allocation 451
452 choices based on how much they would like to spend on the Typing task for the 20- 452
453 minute main session 24 hours before. Figure 5 illustrates how the Time Choices of 453
454 participants were distributed along 5 focal points: The modal action for about a third of 454
455 all participants, was choosing to divide their time equally between the two tasks, then 455
456 another third chose to only type or watch videos, with 15% choosing typing only and 456
457 15% choosing watching only. The remainder of participants were distributed around 457
458 either $1/4$ of the time typing, or $3/4$ of the time typing. Overall, the day 1 decisions can 458
459 be characterized as symmetrical around dividing time across either task. 459

Figure 5: Distribution of Time Choice



Note: This figure shows the distribution of participants' Time Choices for the 20-minute Main Session, measured as the proportion of time planned for typing. The histogram reveals clustering around five focal points: Approximately 20% chose all watching, 12% chose 25% typing, 29% chose equal time allocation, 9% chose 75% typing, and 14% chose all typing. The box plot below shows the overall distribution with median at 0.5.

Based on the observed distribution of the Time Choices, we categorize the day 1 intentions of participants into three distinct groups: Those who almost always planned to type, those who almost always planned to watch videos, and those who planned to mix between the two tasks in various proportions. Table 4 summarizes each group.

Table 4: Participant types by planned time allocation on day 1

	Mostly watching	Alternate	Mostly typing
<i>Time Choice</i>			
Share	28.3%	53.8%	17.9%
Mean (SD)	0.07 (0.08)	0.49 (0.15)	0.95 (0.07)
Median [Min, Max]	0.00 [0.00, 0.23]	0.50 [0.25, 0.75]	1.00 [0.78, 1.00]

Note: This table presents the clustering of participants based on their planned time allocation choices for the 20-minute Main Session on day 1. Watching-focused participants planned to spend 25% or less of their time typing, alternating participants planned to spend between 25% and 75% of their time typing, and typing-focused participants planned to spend more than 75% of their time typing.

4.2 Main Session: Actualized typing decisions on day 2

How did participants allocate their time during the 20-minute main session on day 2? Because we collected second-by-second data on participant behavior, we can track how they allocated their time between the two tasks. Based on the observed distribution of the actualized time spent typing, we categorize the day 2 behavior into three groups following the same rule: Those who almost always typed (more than 75% typing), those who almost always watched videos (less than 25% typing), and those who mixed between the two tasks in various proportions. Table 5 summarizes each type. We observe that the largest share of participants are now either alternating or mostly typing.

Table 5: Participant types by actualized time allocation on day 2

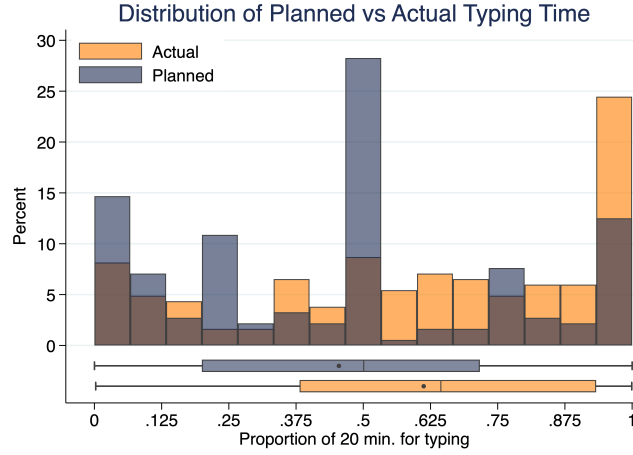
	Mostly watching	Alternate	Mostly typing
<i>Type Time</i>			
Share	19.3%	39.8%	40.9%
Mean (SD)	0.09 (0.08)	0.54 (0.13)	0.94 (0.08)
Median [Min, Max]	0.08 [0.00, 0.24]	0.52 [0.28, 0.74]	0.97 [0.75, 1.00]

Note: This table presents the clustering of participants based on their actualized time allocation during the 20-minute Main Session on day 2. Type Time represents the proportion of time actually spent typing. Mostly watching participants spent 25% or less of their time typing, alternating participants spent between 25% and 75% of their time typing, and mostly typing participants spent more than 75% of their time typing.

4.3 Discrepancy between planned (day 1) and actualized (day 2) time spent on typing

How do participants' day 1 time allocation compare to their actualized time spent on day 2? Participants spent more time on typing than they anticipated, even though we reminded participants of their day 1 plan, and the entire earnings difference between typing for 20 minutes versus watching videos was set to be 60 pence (£0.60). On average, participants planned to allocate 45.5% of their time to typing but actually spent 61.2% of their time on this task, representing an increase of 15.8 percentage points ($t = -8.24$, $p < 0.001$). This extends beyond central tendencies and the entire distribution shifted rightward, with median actual typing time (64.3%) substantially exceeding median planned time (50.0%). The Kolmogorov-Smirnov test confirms that planned and actual distributions differ ($D = 0.86$, $p < 0.001$), indicating systematic deviations from day 1 intentions.

Figure 6: Distribution of Planned vs Actual Time Allocation



Note: This figure compares the distribution of participants' planned time allocation (Day 1) with their actual time spent typing (Day 2), measured as the proportion of the 20-minute session. The rightward shift in the actual distribution demonstrates participants' tendency to spend more time on the effortful typing task than originally planned.

To account for the substantial heterogeneity in both day 1 and day 2 decisions, we compare transitions from participants' day 1 type to their day 2 type in Table 6. We categorize participants into three behavioral types based on their time allocation: mostly watching (< 25% typing), alternate (25-75% typing), and mostly typing (> 75% typing). Overall, 63.0% of participants remained in the same behavioral category between planning and implementation, indicating moderate consistency in behavioral types despite the shift in continuous measures.

Mostly watching and alternate types showed the least adherence to their day 1 plans (59.6% vs 56.6% respectively), with both groups showing upward shifts toward more typing. Among mostly watching planners, 40.4% moved to higher typing categories (30.8% to alternate, 9.6% to mostly typing). Similarly, 40.4% of alternate planners shifted to mostly typing, while only 3.0% moved down to mostly watching. In contrast, mostly typing planners exhibited the highest stability at 87.9%. A chi-square test confirms that deviation rates differ significantly across initial behavioral types ($\chi^2 = 109.00$,

$p < 0.001$), indicating that initial behavioral intentions are strong but heterogeneous predictors of subsequent implementation patterns.

Table 6: Transition matrix: Planned vs actual behavior types

Actual (Day 2)	Planned (Day 1)		
	Mostly watching	Alternate	Mostly typing
Mostly watching	59.6%	3.0%	3.0%
Alternate	30.8%	56.6%	9.1%
Mostly typing	9.6%	40.4%	87.9%

Note: This table shows transitions between planned behavior types (Day 1) and actual behavior types (Day 2). Mostly watching (< 25% typing time), Alternate (25-75% typing time), Mostly typing (> 75% typing time). Diagonal entries show participants who remained consistent (63.0% overall). Off-diagonal entries show deviations: mostly watching planners (40.4% deviated), alternate planners (43.4% deviated), and mostly typing planners (12.1% deviated).

This represents a departure from traditional models of time inconsistency and present bias, which typically predict that individuals will succumb to immediate temptation rather than adhere to effortful plans. Instead, participants demonstrated what might be characterized as ‘reverse present bias’ and systematically allocated more time to the cognitively demanding task than originally intended. The robustness of this finding across the distribution suggests that factors beyond self-control failures govern planning and its implementation in our setting, and potentially include how both tasks and the decision environment were perceived in the experiment.

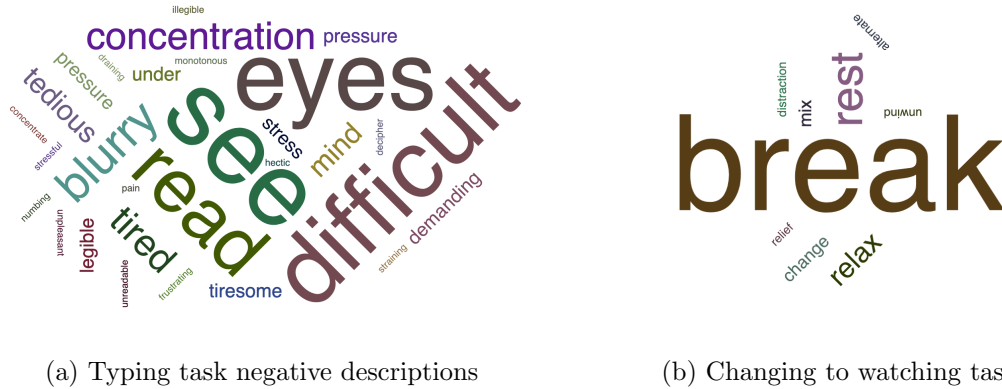
4.4 Experimenter demand effects

Why did participants end up typing more than their day 1 plans? To bridge the gap between our theoretical view of the experimental setup and how participants perceived it, we asked how participants decided to allocate their time during the 20-minute main session once the session was concluded. We explore these by analyzing participants’

515 qualitative answers to an open-ended question on how they decided to divide their time 515
 516 between the two tasks using dictionary-based concept detection methods reviewed by 516
 517 [Ash and Hansen \(2023\)](#). 517

518 We employ dictionary-based concept detection methods within a bag-of-words frame- 518
 519 work to analyze participants’ qualitative responses. Following standard approaches in 519
 520 computational text analysis, we preprocess the text data by converting to lowercase and 520
 521 apply pattern matching to identify specific term sets that relate to negative perceptions 521
 522 of the typing task and breaks towards the watching task. Using regex pattern matching, 522
 523 we create binary indicators for documents containing predetermined words associated 523
 524 with task difficulty (e.g., “tiresome”, “frustrating”, “difficult”) and break-taking behavior 524
 525 (e.g., “break”, “rest”, “relax”) - with the exception of “switch” which typically described 525
 526 simple task changing behavior without the implication of taking a break from typing. 526
 527 This approach allows quantifying the prevalence of common conceptual themes across 527
 528 participants’ obligatory open-ended responses. We match algorithmic output with a 528
 529 manual review to confirm our results about participant perceptions. Figures 7a and 7b 529
 530 summarize these words by their frequency. 530

Figure 7: Word clouds for participant perceptions



Note: Panel (a) shows words describing negative perceptions of the typing task. Panel (b) shows words describing switching to the watching task as break-taking behavior. Word size corresponds to frequency of mention across responses.

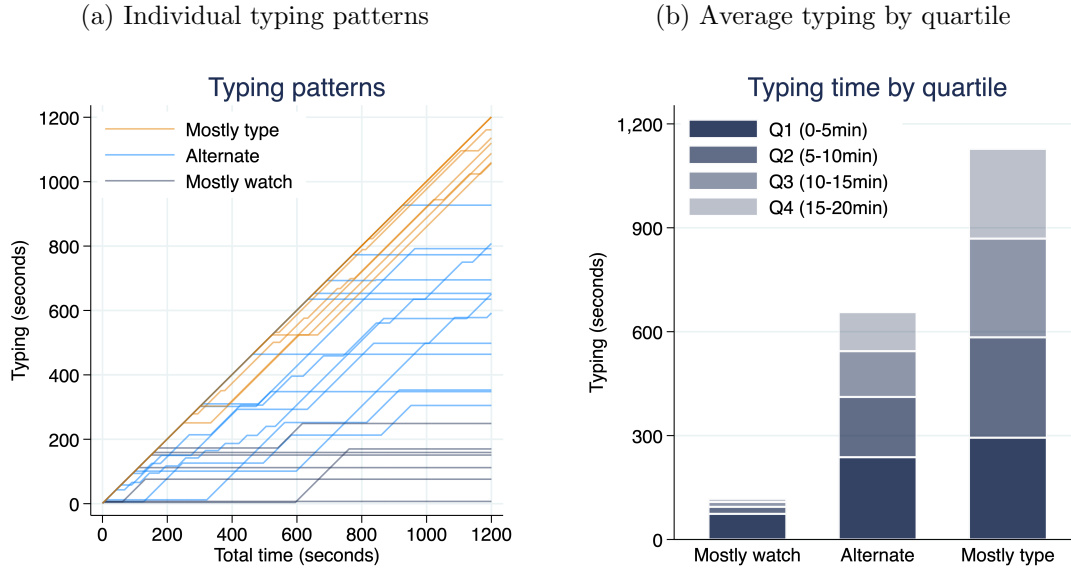
Our mixed methods analysis reveals that when asked to describe how they decided to switch between the tasks, 29% of participants described the typing task negatively at their own will, i.e., without being asked to comment on the typing task itself, while 23% specifically described watching videos as a break from the typing task. These findings suggest that the decision environment was perceived by many as a work environment, where one should type as much as possible while taking as few breaks as possible. The observed increases across all but always typing types, along with the qualitative perceptions suggest that experimenter demand effects have effected participant behavior.

4.5 Effect of achieving day 1 typing objective

One test for this hypothesis is looking at what participants do around the window of achieving their day 1 goals. If their behavior changes, i.e., they start typing less after achieving their goal, this would indicate adherence to day 1 goals and limited distortion due to experimenter demand effects. Should they continue typing as if nothing happened, we should be worried of severe problems in the desing of the experiment. We test this exploratory hypothesis in the next section.

We categorize participants in their actualized typing time on day 2 using the same three categories from day 1 (see Figure 8a). For participants who are “alternating”, typing periods are interrupted by watching periods, following with the “taking breaks” analogy made by participants. We observe substantial individual heterogeneity in lengths of typing and watching periods, and divide the main session across 5-minute periods to understand typing to watching proportions of participants. We find that typing periods get shorter and shorter, suggesting changing between tasks may be strategic as participants get tired of typing or achieve their day 1 goals (see Figure 8b).

Figure 8: Time spent typing



Note: Panel (a) shows typing time for 30 random participants across the 20-minute session. Panel (b) shows average typing time by 5-minute periods, revealing declining typing duration over time.

Why do participants reduce time spent typing as they advance? If they are typing without regard to their day 1 objective, their typing shouldn't be affected by attaining their day 1 goal, and reduce gradually as fatigue or frustration grows from doing the typing task. Instead, if they care about hitting their goal, we should see a sharp drop in time spent typing right after attaining it.

To examine behavioral changes as participants achieve their day 1 plans, we implement a sharp regression discontinuity design using the moment participants reach their planned typing duration as the cutoff. We aggregate binary task choices (whether typing or watching) into 30-, 60-, and 90-second bins around this threshold and estimate the discontinuous change in the probability of choosing the typing task. Results in Table 7 reveal a significant negative effect immediately after plan achievement: participants are 27.4 percentage points less likely to continue typing in the 30-second window following plan completion (robust estimate: -0.274 , $p = 0.007$) and 25.5 percentage points less

likely in the 60-second window (robust estimate: -0.255, $p = 0.003$). However, this effect dissipates over longer horizons, becoming statistically insignificant in the 90-second window (robust estimate: -0.075, $p = 0.487$). These findings suggest that achieving initial plans triggers an immediate behavioral shift toward leisure consumption, consistent with participants treating plan completion as a license to reduce effort temporarily.

Table 7: Regression Discontinuity estimates on typing after achieving day 1 plans

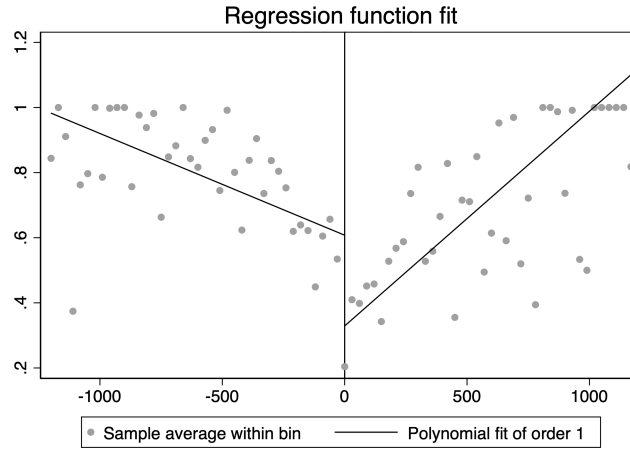
	Time Window		
	30-seconds	60-seconds	90-seconds
Proportion typing	-0.274*** (0.101)	-0.255*** (0.087)	-0.075 (0.108)
Obs. (left)	66,786	67,985	67,119
Obs. (right)	80,270	89,006	92,999
Clusters (left)	157	154	155
Clusters (right)	154	157	161
Bandwidth	339.9	368.3	422.3

Note: This table presents robust regression discontinuity estimates of the treatment effect on task choice at the moment participants achieve their day 1 planned typing duration. Standard errors clustered by participant in parentheses. The outcome variable is a binary indicator for choosing the typing task. Running variables are binned at 30-, 60-, and 90-second intervals around the achievement threshold. All estimates use triangular kernel with MSE-optimal bandwidth selection. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Visual inspection along with the results from 90-second window suggests this discontinuity it not all there is to know about participant behavior: While there is an initial drop in typing rate after achieving day 1 plans, participants eventually ramp up the time they spend at typing (see Figure 9). The positive slope on the right hand side illustrates this effect. Participants who achieve their day 1 plans earlier may in fact eventually get

“bored” of the watching task as well - a behavior that shows how our efforts to create an addictive leisure task may have ultimately failed. These observations are coming from the about at least a third of all participants who were mostly watchers or alternaters in their day 1 types, but have increased their typing task consumption on day 2 (about respectively $(184 * 0.28 * 0.4)/184 = 11,2\%$ and $(184 * 0.54 * 0.4)/184 = 21,6\%$ of participants).

Figure 9: Linear fit for typing rate before and after achieving day 1 plan



Note: This figure plots the average typing rate across 30-second bins for participants as they approach their day 1 goal. The negative slope on the left hand side indicates participants working less and less as they approach the cutoff. The drop showcases the discontinuity where achieving the day 1 goal causes in typing behavior. The positive slope on the right hand side reveals another trend: As time goes on, participants tend to go back to the typing task and decrease their video consumption. This demonstrates participants’ tendency to spend more time on the effortful typing task than originally planned.

4.6 Does Autoplay increase video consumption?

As participants were mostly focused on typing, they did paid less attention to the watching task, i.e, reducing its consumption value as evidenced by day 2 behavior. Consequently, in our setting experimenter demand effects is acting as a confound, in the sense

of making more difficult to show statistically significant evidence in support of the hypotheses regarding the direction of deviations from the day 1 plans and the effects of the **Autoplay** condition. A lack of evidence in support of these objectives in our setting could be due not to a genuine failure of the corresponding hypotheses but rather to experimenter demand effects (Zizzo, 2010). With these in mind, we summarize variables of interest from the main 20-minute session across both conditions in Table 8. Two sample t -tests show no significant differences in any of the outcome variables. This suggests that **Autoplay** condition had no impact on the watching behavior observed in the experiment. Further, as established before, participants spend more time typing when compared to their Time Choice on day 1 ($0.158, t(183) = 8.92, p < 0.001$).

Table 8: Sample statistics for variables of interest

	Autoplay	Control	<i>p</i>
	N = 79	N = 105	[A = C]
<i>Time Choice</i>			
Mean (SD)	0.48 (0.31)	0.44 (0.33)	0.410
Median [Min, Max]	0.50 [0.00, 1.00]	0.50 [0.00, 1.00]	
<i>Proportion Typing</i>			
Mean (SD)	0.65 (0.31)	0.58 (0.33)	0.173
Median [Min, Max]	0.72 [0.00, 1.00]	0.56 [0.00, 1.00]	
<i>Nb. of sessions</i>			
Mean (SD)	5.76 (4.98)	4.89 (5.09)	0.246
Median [Min, Max]	4.00 [1.00, 18.00]	3.00 [1.00, 28.00]	
<i>Session length</i>			
Mean (SD)	456.11 (383.22)	546.35 (411.53)	0.131
Median [Min, Max]	314.75 [68.17, 1212.00]	418.67 [44.93, 1314.00]	
<i>Nb. of CAPTCHA submissions</i>			
Mean (SD)	19.76 (12.11)	17.21 (11.62)	0.150
Median [Min, Max]	19.00 [0.00, 53.00]	15.00 [0.00, 45.00]	
<i>Nb. of videos watched</i>			
Mean (SD)	31.33 (26.34)	32.72 (26.39)	0.723
Median [Min, Max]	25.00 [0.00, 80.00]	34.00 [0.00, 77.00]	
<i>Content rating</i>			
Mean (SD)	6.96 (2.18)	6.62 (2.45)	0.327
Median [Min, Max]	7.00 [2.00, 10.00]	7.00 [0.00, 10.00]	

Note: This table presents key outcome variables from the Main Session. “Time Choice” refers to participants’ stated preferences for typing on day 1 (proportion), while “Proportion Typing” represents the actual proportion of time spent on typing task during the 20-minute Main Session on day 2. Nb. of sessions is the number of typing or watching sessions participants completed and Session length represents the mean duration of individual sessions in seconds. Content rating reflects participants’ evaluation of video content on a 0-10 scale. *p*-values are from two-sample *t*-tests comparing experimental conditions.

597 To formally test [Hypothesis 1](#), i.e., effects of the **Autoplay** condition on time spent 597
598 on typing, we run the regression specified in Equation 1. Our analysis reveals that the 598
599 autoplay manipulation had no statistically significant effect on actual typing behavior. 599
600 Table 9 presents results for typing proportion as the dependent variable. The coefficient 600
601 on the autoplay treatment ranges from 0.12 to 0.20 across specifications but remains 601
602 statistically insignificant in all models (p-values > 0.16). However, participants' day 602
603 1 plans emerge as a powerful predictor of day 2 behavior: a one standard deviation 603
604 increase in planned typing time is associated with a 0.67 standard deviation increase in 604
605 actual typing time (p < 0.001). Including day 1 plans increases the model's explanatory 605
606 power dramatically, with R-squared rising from 1% to 46%. 606

Table 9: Effect of Autoplay Treatment on Typing Behavior

	Proportion Typing (z-score)		
	(1)	(2)	(3)
Autoplay	0.203 (0.147)	0.121 (0.109)	0.119 (0.110)
Time Choice (z-score)		0.673*** (0.057)	0.651*** (0.081)
Autoplay × Time Choice			0.057 (0.110)
Constant	−0.087 (0.100)	−0.052 (0.076)	−0.053 (0.076)
Obs.	184	184	184
R-squared	0.010	0.462	0.463
F-statistic	1.91	77.42	61.52

Note: Robust standard errors in parentheses. All variables are standardized. Time Choice represents participants' planned typing proportion from day 1. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Similarly, Table 10 examines the effect on video consumption, directly testing our primary hypothesis that autoplay would increase video watching. Across all specifications, the autoplay coefficient is statistically insignificant and even negative in the baseline model (-0.053 , $p = 0.723$). This finding contradicts our theoretical prediction and suggests that the autoplay feature did not meaningfully alter participants' video consumption patterns. Again, day 1 plans prove to be the dominant predictor: participants who planned more typing watched significantly fewer videos ($\beta = -0.66$, $p < 0.001$), explaining 44% of the variance in video consumption.

Table 10: Effect of Autoplay Treatment on Video Watching Behavior

	Nb. of Videos Watched (z-score)		
	(1)	(2)	(3)
Autoplay	-0.053 (0.149)	0.029 (0.112)	0.031 (0.112)
Time Choice (z-score)		-0.664^{***} (0.056)	-0.619^{***} (0.079)
Autoplay $\times$ Time Choice			-0.114 (0.105)
Constant	0.023 (0.098)	-0.012 (0.076)	-0.010 (0.076)
Obs.	184	184	184
R-squared	0.001	0.440	0.443
F-statistic	0.13	73.06	61.53

Note: Robust standard errors in parentheses. All variables are standardized. Time Choice represents participants' planned typing proportion from day 1. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

For robustness, we provide the regression outputs from the unrestricted dataset in Appendix Table A.2 and A.3. Comparing the interaction models (column 3), we observe

that the autoplay treatment effect on typing behavior decreases from 0.119 in the restricted sample to 0.055 in the unrestricted sample, while the autoplay treatment effect on video watching increases from 0.031 in the restricted sample to 0.132 in the unrestricted sample. However, the null results remain unchanged across both specifications. Because we cannot be sure these 40 participants actually engaged in watching videos, we base our conclusions on the restricted sample of 184.

To summarize, we observe two key patterns in the data that help explain our null findings. First, we observe contradictory results: the **Autoplay** condition increases time spent typing by 0.12 standard deviations (though statistically insignificant) while simultaneously increasing the number of videos watched by 0.03 standard deviations (also insignificant). If people type more, shouldn't they watch fewer videos? We identified a mechanical confound in our experimental design: the total time lost during video transitions. Participants in the **Control** condition, who had to manually click to start each new video, lost an average of 46.52 seconds due to these transition delays whereas those in the **Autoplay** condition lost only 1.08 seconds. This large difference of 45.45 seconds per session is statistically significant ($t(105) = 4.89, p < 0.001$). We find that this difference in time lost not watching videos plausibly explain the contradicting results and underscores the importance of looking at the unbiased outcome variable (i.e., number of videos watched) when evaluating the effects of the autoplay feature.

Second, assuming the actual effect size for the number of videos watched lie somewhere between the restricted and unrestricted samples, our power calculations based on pretests were insufficient: The required sample size to provide a two-sided test for [Hypothesis 1b](#) with the usual parameter values ($\alpha = .05, \beta = .2$) would be between 300 to 5000 per treatment, with common standard deviation as the root MSE from the standardized number of videos watched variable in the third regression. Other solutions to this power problem are design-related and involve extending the main-session duration to above 20 minutes, or nudging participants to mix more between tasks to avoid corner solutions (i.e., higher earnings when setting Time Choice between .25 and .75).

Table 11: Total Transition Time by Treatment Condition

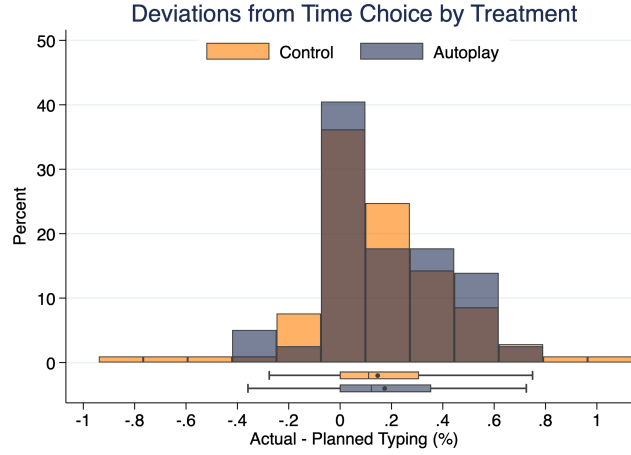
	Control	Autoplay	Difference
Total time lost	46.52 (9.27)	1.08 (0.65)	45.45*** (9.29)
Observations	105	79	184
<i>t</i> -statistic			4.89

Note: Standard errors are in parentheses. Time is measured in total seconds lost to video transitions per session. The difference was tested using a two-sample *t*-test with unequal variances. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

4.7 Effects of autoplay on deviations from Time Choice

Does the **Autoplay** condition have an effect on deviations from day 1 Time Choice? To assess [Hypothesis 2](#), we calculated the deviation by subtracting participants' planned typing proportion from their actual typing proportion (actual minus planned). Figure 10 shows the distribution of deviations from Time Choice across treatments, which appears right-skewed with positive values indicating participants typed more than originally planned. Both treatment groups systematically over-typed relative to their day 1 plans: Participants in the **Control** condition deviated by an average of 14.6 percentage points (SD = 27.6) while Autoplay participants deviated by 17.3 percentage points (SD = 23.6). However, a two-sample *t*-test reveals no statistically significant difference between means ($t = -0.694$, $p = 0.488$). The Kolmogorov-Smirnov test similarly finds no significant difference in distributions across conditions ($D = 0.085$, $p = 0.903$). These results indicate that while participants in both conditions systematically spent more time on the effortful typing task than they initially planned, the **Autoplay** condition did not significantly alter the magnitude or pattern of these deviations. Contrary to our hypothesis, we find no evidence that autoplay led participants to deviate more toward video watching behavior.

Figure 10: Distribution of Deviations from Time Choice by Treatment



Note: This figure shows the distribution of deviations from participants' day 1 time allocation plans, calculated as actual typing proportion minus planned typing proportion. Positive values indicate participants typed more than planned, while negative values indicate less typing than planned. The distributions are similar across treatment conditions, with both groups showing systematic over-typing relative to initial plans.

4.8 WTP for the commitment device (H3)

[Hypothesis 3](#) explores whether participants would choose to turn Autoplay off when given the possibility, implying positive demand for a commitment device. We assess this demand by comparing the share of participants choosing to turn Autoplay on or off along with the associated bonus payments for each of the nine decisions in the price list. To quantify participants' willingness to pay (WTP) for Autoplay, we estimate the regression specification in Equation 2, where the dependent variable is the binary choice of selecting Autoplay (1) or turning it off (0), and the independent variable is the bonus amount in pounds from the MPL. The model is clustered at the participant level to account for 9 repeated decisions by the same individual.

Results are presented in Table 12. The coefficient on the bonus amount is 27.97 ($p < 0.001$), indicating that participants are highly sensitive to the financial cost of maintaining Autoplay. The WTP for Autoplay can be calculated as the negative ratio of the

constant to the bonus coefficient: $-0.6263/27.9729 = -0.0224$ pounds, or approximately 2.24 pence. This suggests that participants value Autoplay at 2.24 pence on average and would be willing to forgo this amount to maintain access to the feature rather than have it turned off. In light of how participants operated in the decision-making environment, it appears that participants did not perceive a tradeoff from Autoplay in terms of increasing video consumption, but instead viewed it as facilitating leisure consumption before returning to the typing task.

Table 12: Willingness to Pay for Autoplay

	(1)
Bonus (£)	27.973*** (6.277)
Constant	0.626** (0.199)
Obs.	468
Clusters	52
Pseudo R ²	0.658
Wald $\chi^2(1)$	19.86

Note: Logistic regression with standard errors clustered at the participant level in parentheses. Dependent variable is binary choice of selecting Autoplay (1) or turning it off (0). Bonus amounts range from -0.5 to 0.5 pounds. The pseudo R-squared (0.6584) indicates that the bonus amount explains a substantial portion of the variation in participants' choices. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

5 Discussion - integrate to introduction, redo and send

6 Conclusion

EXPERIMENTAL DEMAND EFFECTS AND SOCIAL IMAGE CONCERNS. THE
EXPERIMENT IS SET AS A WORKING ENVIRONMENT. PARTICIPANTS PER-

CEIVED THE DECISION AS ONE TO WORK AS MUCH AS POSSIBLE.

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777 **A Additional Figures and Tables**

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778 **Additional Figures**

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Table A.1: Comparison of main sample vs flagged participants

	Main Sample N = 184	Flagged Sample N = 40	<i>p</i> [M = F]
<i>Gender</i>			
Female	89 (48.37%)	13 (32.50%)	0.068
<i>Age</i>			
Mean (SD)	42.29 (11.41)	42.85 (10.75)	0.768
<i>Employment</i>			
Employed (full time)	102 (55.43%)	27 (67.50%)	0.162
Employed (part time/self)	48 (26.09%)	8 (20.00%)	0.420
Not employed	34 (18.48%)	5 (12.50%)	0.366
<i>Income</i>			
Low (0-15)	56 (30.43%)	6 (15.00%)	0.048*
Middle (15-50)	103 (55.98%)	31 (77.50%)	0.012*
High (50+)	25 (13.59%)	3 (7.50%)	0.291
<i>Marital Status</i>			
Married/Partnership	95 (51.63%)	21 (52.50%)	0.921
Single	77 (41.85%)	17 (42.50%)	0.940
Previously married	12 (6.52%)	2 (5.00%)	0.719
<i>Treatment Assignment</i>			
Autoplay condition	79 (42.93%)	25 (62.50%)	0.025*

Note: This table compares demographic characteristics between the main sample (N=184) and participants flagged for potential inattention. Flagged participants reported engaging with videos but had mouse activity detected outside the experiment window, making it impossible to verify genuine engagement. *p*-values for binary outcomes are from two-sample tests of proportions; for continuous variables, from two-sample *t*-tests. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A.2: Effect of Autoplay Treatment on Typing Behavior

	Proportion Typing (z-score)		
	(1)	(2)	(3)
Autoplay	0.136 (0.133)	0.055 (0.108)	0.055 (0.108)
Time Choice (z-score)		0.595*** (0.061)	0.580*** (0.089)
Autoplay $\times$ Time Choice			0.034 (0.119)
Constant	-0.063 (0.094)	-0.026 (0.078)	-0.027 (0.079)
Obs.	224	224	224
R-squared	0.005	0.357	0.357
F-statistic	1.04	50.28	35.00

Note: Robust standard errors in parentheses. All variables are standardized. Time Choice represents participants' planned typing proportion from day 1. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Table A.3: Effect of Autoplay Treatment on Video Watching Behavior

	Nb. of Videos Watched (z-score)		
	(1)	(2)	(3)
Autoplay	0.051 (0.135)	0.131 (0.109)	0.132 (0.109)
Time Choice (z-score)		-0.590*** (0.060)	-0.555*** (0.084)
Autoplay $\times$ Time Choice			-0.077 (0.118)
Constant	-0.023 (0.090)	-0.061 (0.075)	-0.059 (0.075)
Obs.	224	224	224
R-squared	0.001	0.347	0.348
F-statistic	0.14	48.91	34.21

Note: Robust standard errors in parentheses. All variables are standardized. Time Choice represents participants' planned typing proportion from day 1. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.